

An Introduction to 5G Technology

Even as 4G mobile telecommunication standards slowly gain in popularity, there is already talk of the fifth-generation or 5G standards that will further enhance communications on the move

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5G mobile telecommunication standards stand for fifth-generation advancements made in the mobile communications field. These comprise packet switched wireless systems using orthogonal frequency division multiplexing (OFDM) with wide area coverage, high through-

put at millimetre waves (10 mm to 1 mm) covering a frequency range of 30 GHz to 300 GHz, and enabling a 20 Mbps data rate to distances up to 2 km. The millimetre-wave band is the most effective solution to the recent surge in wireless Internet usage. These specifications are capable of providing 'wireless world wide web' (WWWW) applications.

The WWWW allows a highly flexible network (flexible channel bandwidth between 5 and 20 MHz, optimally up to 40 MHz), and dynamic ad-hoc wireless network (DAWN). This technique employs intelligent antennae (e.g., switched beam antennae and adaptive array antennae) and the flexible modulation method, which helps in obtaining bidirectional high bandwidth, i.e., transfer of a large volume of broadcasting data in giga bytes, sustaining more than 60,000 connections and providing 25 Mbps connectivity.

Users of 5G technology can download an entire film to their tablets or laptops, including 3D movies; they can download games and avail of remote medical services. With the advent of 5G, Piconet and Bluetooth technologies will become outdated. The 5G mobile phones would be akin

to tablet PCs, where you could watch TV channels at HD clarity without any interruption.

Key concepts in 5G technology

Future mobile devices equipped with 5G technology will have:

1. Wearable devices with artificial intelligence (AI)
2. Internet Protocol version 6 where the IP address is assigned according to location and the connected network.
3. The ability to connect the user to different wireless access technologies, like 2.5G, 3G, 4G or 5G mobile networks, as well as Wi-Fi and WPAN (wireless personal area network)—or even any other technology to be developed in the future. This is basically a concurrent data transfer path technique.
4. Smart radio. In order to share the same spectrum efficiently during a wireless transmission scheme, the system will adaptively find (search) unused spectrum. This dynamic radio resource management will be achieved in a distributed fashion and rely on software defined transmission.
5. High altitude stratospheric platform station (HAPS) system. This is based on beam division multiple access (BDMA) and group relay techniques.

5G hardware

Ultra wideband networks (UWB). It is already known that Wi-Fi, Wi-Max and cellular wide area communications are long-range radio technologies. But systems like WPAN need short-range radio technology, which helps in achieving higher bandwidths (around 4000 Mbps) but at low energy levels (UWB network) for relaying data from host

